



[PROJECT TITLE GOES HERE]

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Database Systems Engineering and Distributed Backend Development (25CS1302E) | KLH Bachupally Campus, CSE | PBL 2026-27

Abstract

- CKD SmartCare is a DBMS-based patient management system designed for patients, doctors, staff, and administrators to securely manage patient, appointment, medical, and laboratory records. It uses PostgreSQL/MongoDB, FastAPI, SQL queries, and role-based access control (RBAC) for efficient data storage, retrieval, and management.

Problem Statement & Objectives

Problem: Fragmented CKD records make patient data difficult to manage and access.

Objective 1: Centralize patient and medical data.

Objective 2: Provide secure role-based access.

Objective 3: Support CKD risk prediction and monitoring.

Scope: Admin, Doctor, Staff & Patient modules with patient, medical, laboratory, appointment and prediction records.

Tools & Technologies

Front-End: HTML, CSS, JavaScript

Back-End: FastAPI + REST APIs

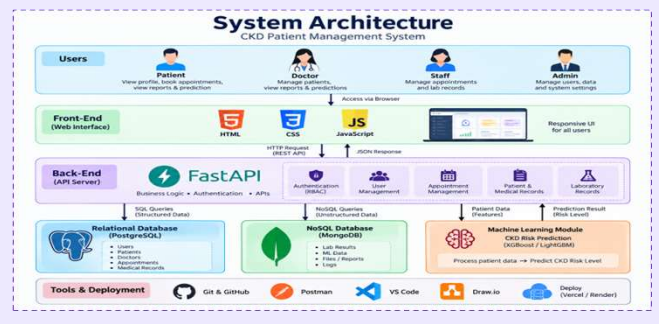
Database: PostgreSQL + MongoDB

Machine Learning: Python, XGBoost, LightGBM

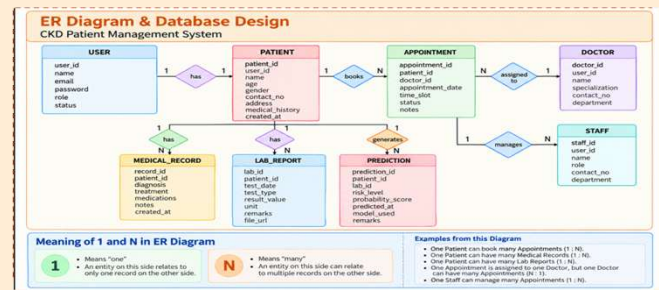
Tools: Git & GitHub, Postman, VS Code, Draw.io

Authentication: JWT / Role-Based Access Control (RBAC)

System Architecture



ER Diagram & Database Design



Feasibility Study

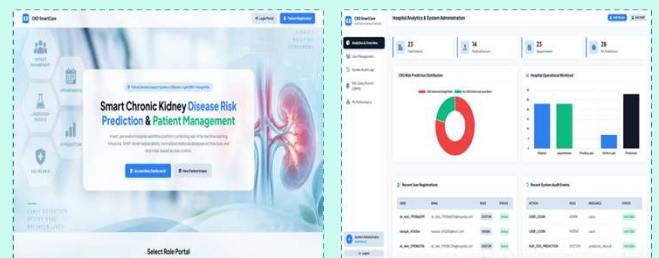
Technical: Uses available skills and open-source technologies like HTML, CSS, JavaScript, FastAPI, PostgreSQL, MongoDB, and Python.

Economic: Low cost using open-source software and free-tier cloud services.

Operational: Simple role-based interface with easy maintenance and use.

Schedule: Can be completed within the semester timeline.

Implementation Screenshots



Results & Testing

Key Outcomes: Successfully manages CKD patient, medical, laboratory, appointment, and prediction records.

CRUD: Create, read, update, and delete operations work correctly.

Validation: Validates user inputs and required fields.

Performance: Provides fast database access and responsive system operations.

Conclusion

What was achieved: Developed a centralized CKD patient management and risk-prediction system.

Objectives Met: Successfully supports patient data management, role-based access, CRUD operations, and CKD risk prediction.

Future Scope

Enhancement 1: Add real-time health monitoring and advanced CKD prediction.

Enhancement 2: Develop a mobile application with cloud-based access.

